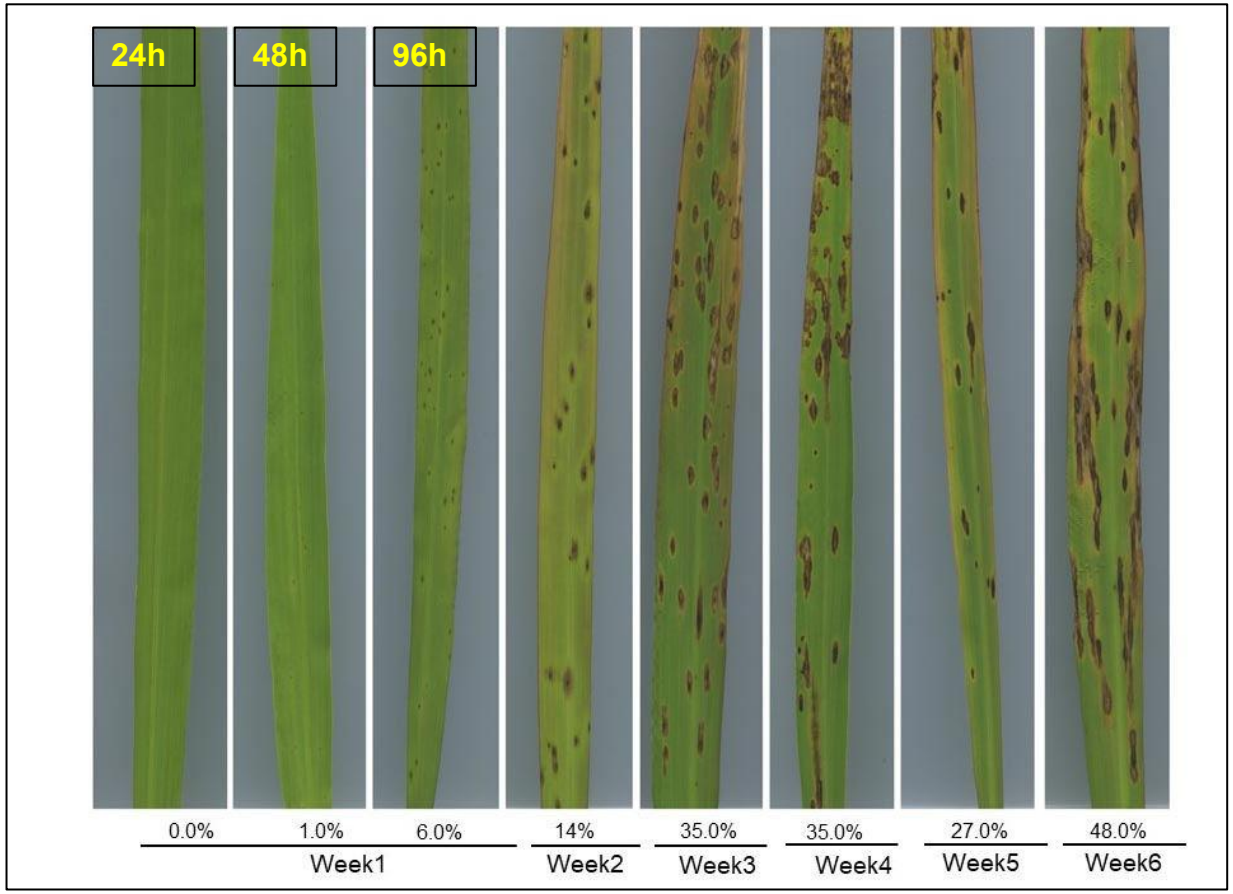
Examples of wild rice diseases

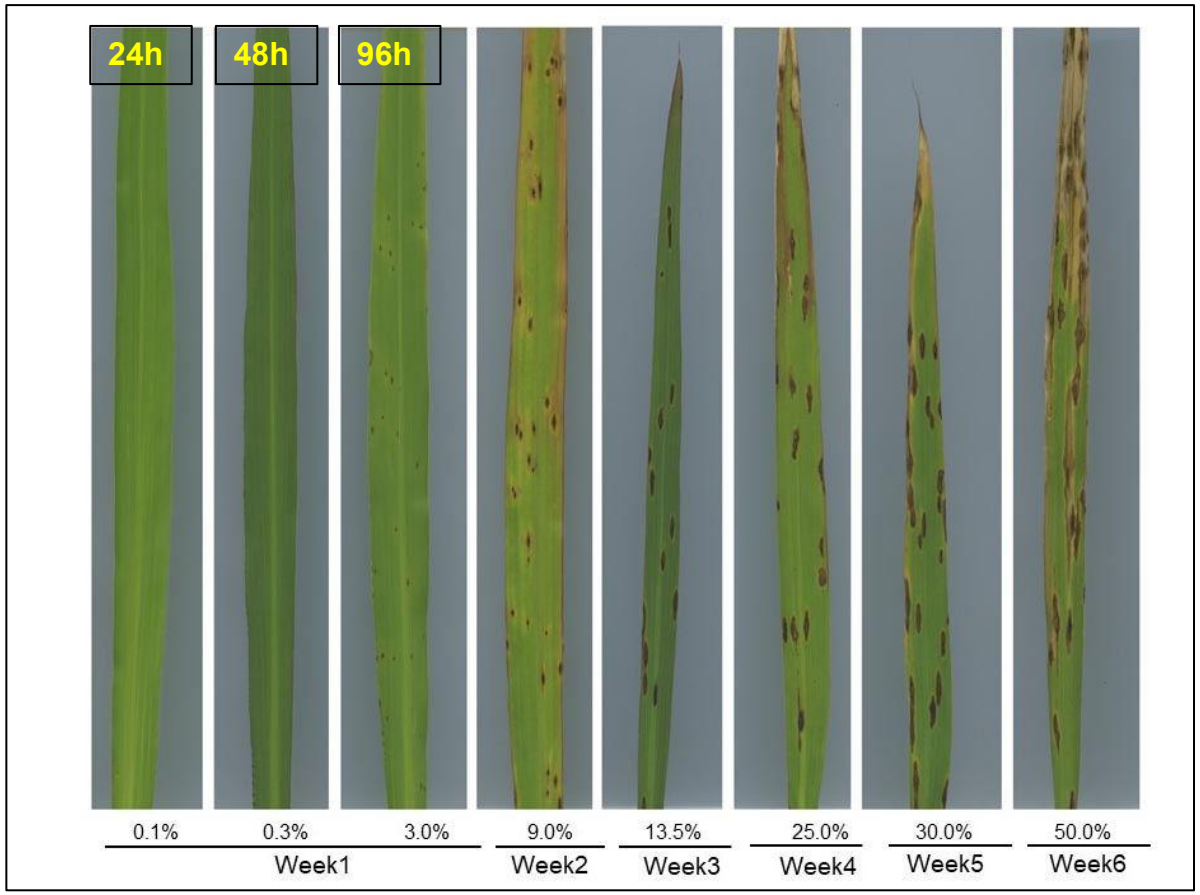
Claudia Castell-Miller

Figure 1a. Standard Analytic Scales for Fungal Brown Spot (artificial inoculation with *Bipolaris oryzae*)

Franklin



Itasca – C12_C3R



Representative leaf (Fl-1) images of disease severity (%) in two wild rice cultivars over time

Author images: Claudia Castell-Miller

Figure 1b. Fungal Brown Spot (natural infection by *Bipolaris oryzae*)

Leaves



Sheath/Stem



Peduncle



Author images: Claudia Castell-Miller

Figure 1c. Fungal Brown Spot (artificial inoculation with *Bipolaris oryzae*)



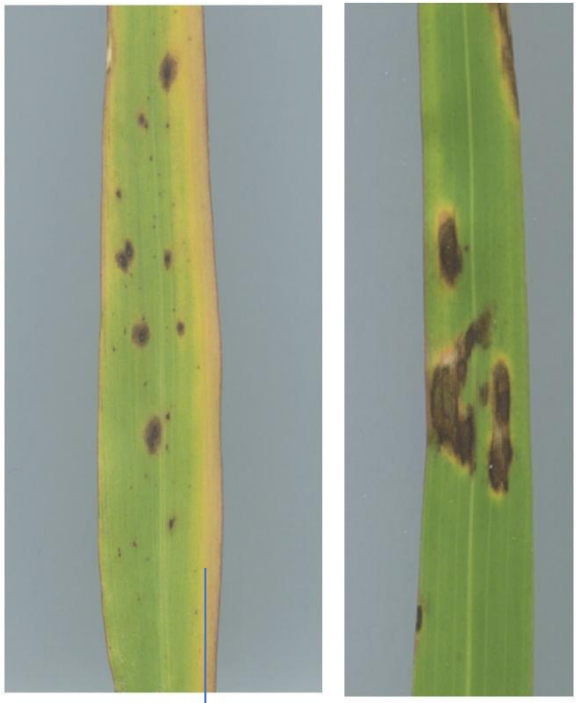
Author images: Claudia Castell-Miller

Figure 2a. Spot blotch scale (*Bipolaris sorokiniana*)



Itasca C-12 C2S (susceptible)

Typical symptoms



Senescence

Author images: Claudia Castell-Miller

Figure 2b. Spot Blotch caused by *Bipolaris sorokiniana*

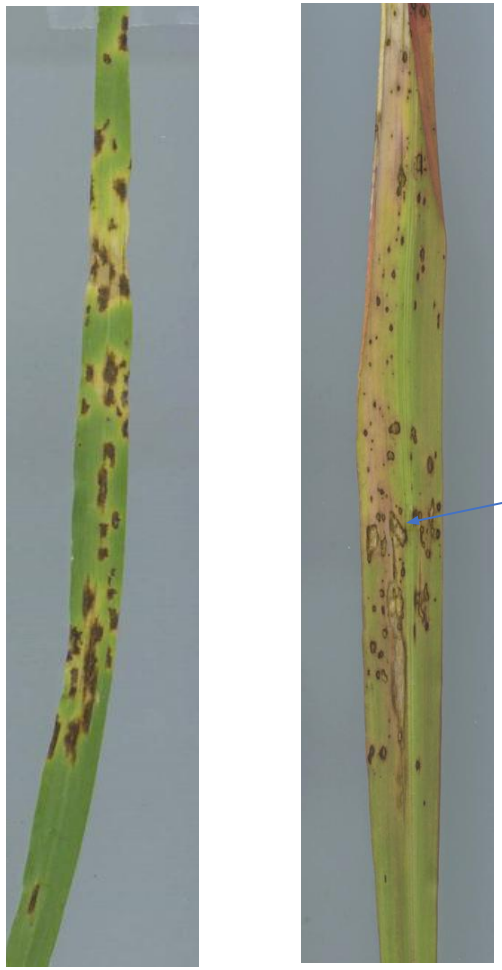
Symptoms after inoculation in greenhouse

Early lesions



Note: Please note that they shape of the lesions are more oval or of even irregular shape than those of fungal brown spot, and they can develop chlorosis; they also coalescence and can become tan (dry).

Fully developed lesions



Lesions in an older leaf (note the orange redness color in the edges and tip may be due to the disease or a combination of disease and due to senescence)

These type of lesions are uncommon

Author images: Claudia Castell-Miller

Figure 2c. Spot Blotch caused by *Bipolaris sorokiniana*



Symptoms after artificial inoculation in greenhouse

Author images: Claudia Castell-Miller

Figure 3a. Zonate eyespot caused by *Bipolaris gigantea*

Field images

Early symptoms





Mature symptoms

Note: Please note that lesions appear as water soak areas (first image) and then they look like “eyes” due to secondary infections. The fungus is found in paddies

Author images: Claudia Castell-Miller

Figure 2b. Zonate eye spot: early lesion stages

No borders



Greenish areas (no borders) around of mature- and no-mature lesions



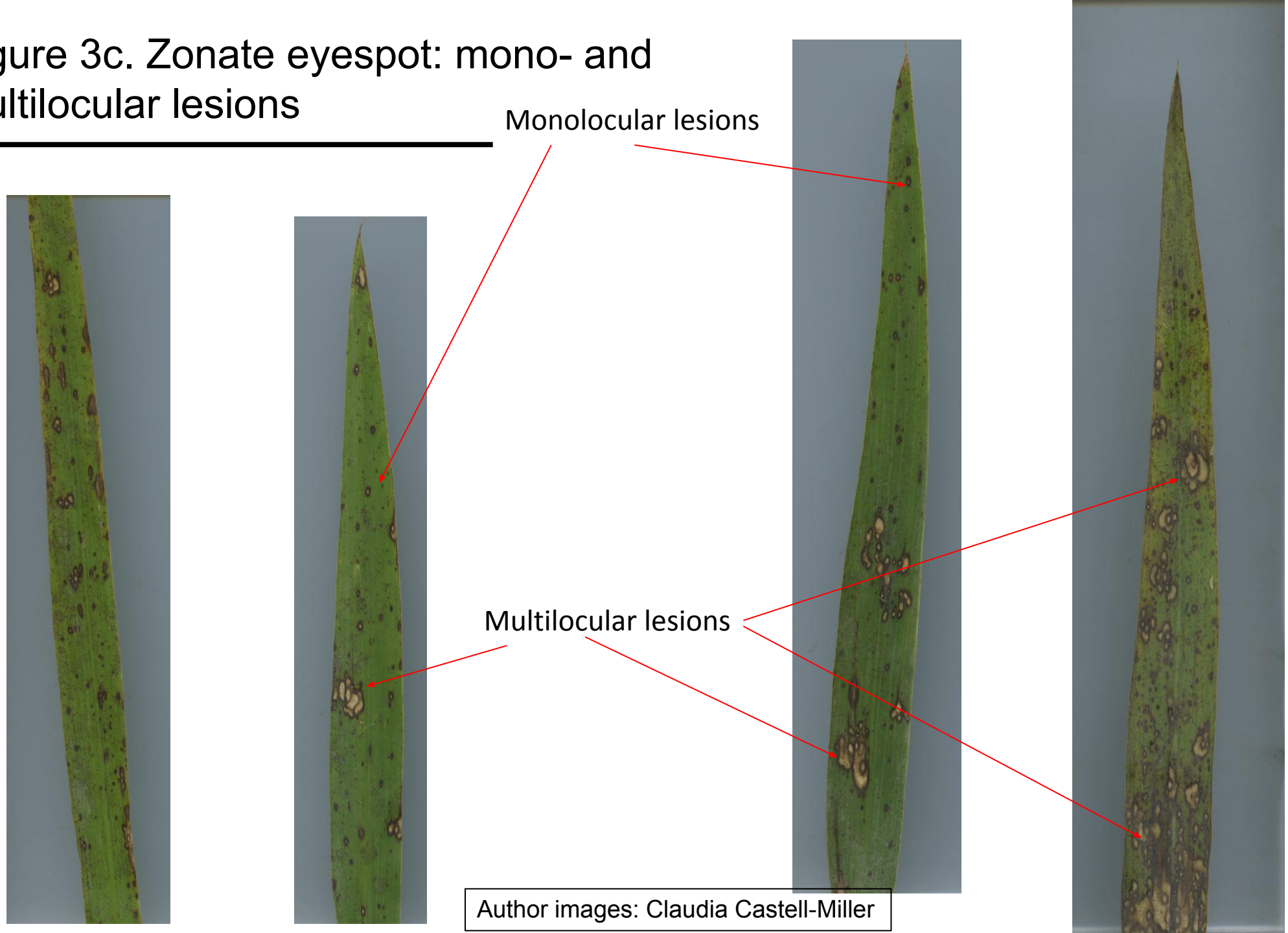
Transition to mature lesions when borders start or are developed





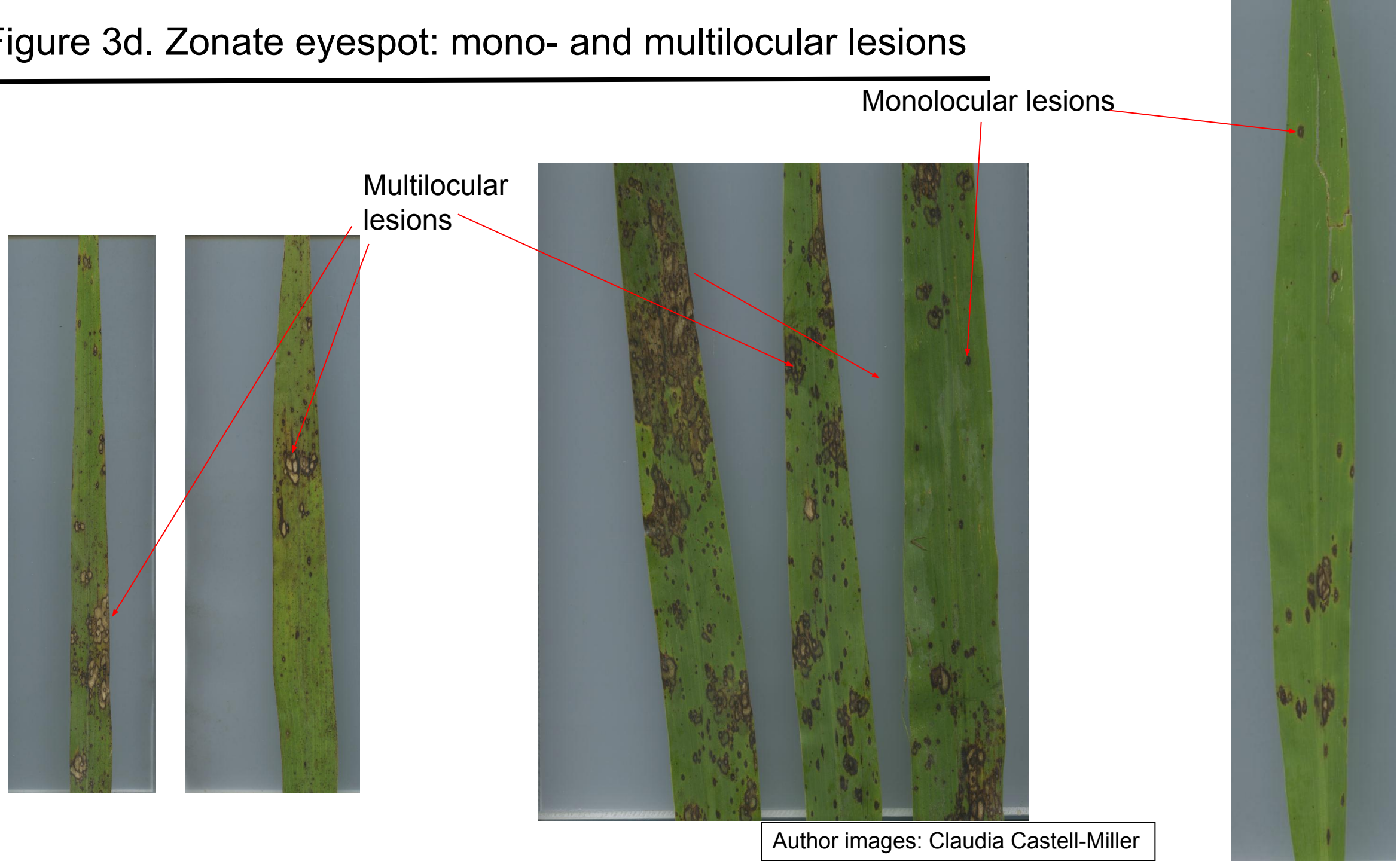
Author images: Claudia Castell-Miller

Figure 3c. Zonate eyespot: mono- and multilocular lesions



Author images: Claudia Castell-Miller

Figure 3d. Zonate eyespot: mono- and multilocular lesions

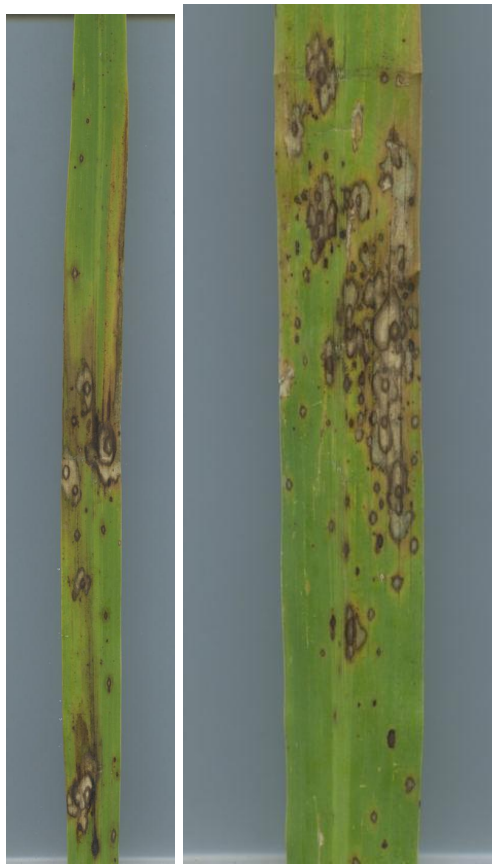


Author images: Claudia Castell-Miller

Figure 3e. Zonate eyespot caused by *B. gigantea* showing chlorosis/discoloration around mature lesions











Author images: Claudia Castell-Miller

Figure 3f. Zonate eyespot caused by *B. gigantea* showing symptoms on the edge (1) and on a dead leaf (2)

(1)



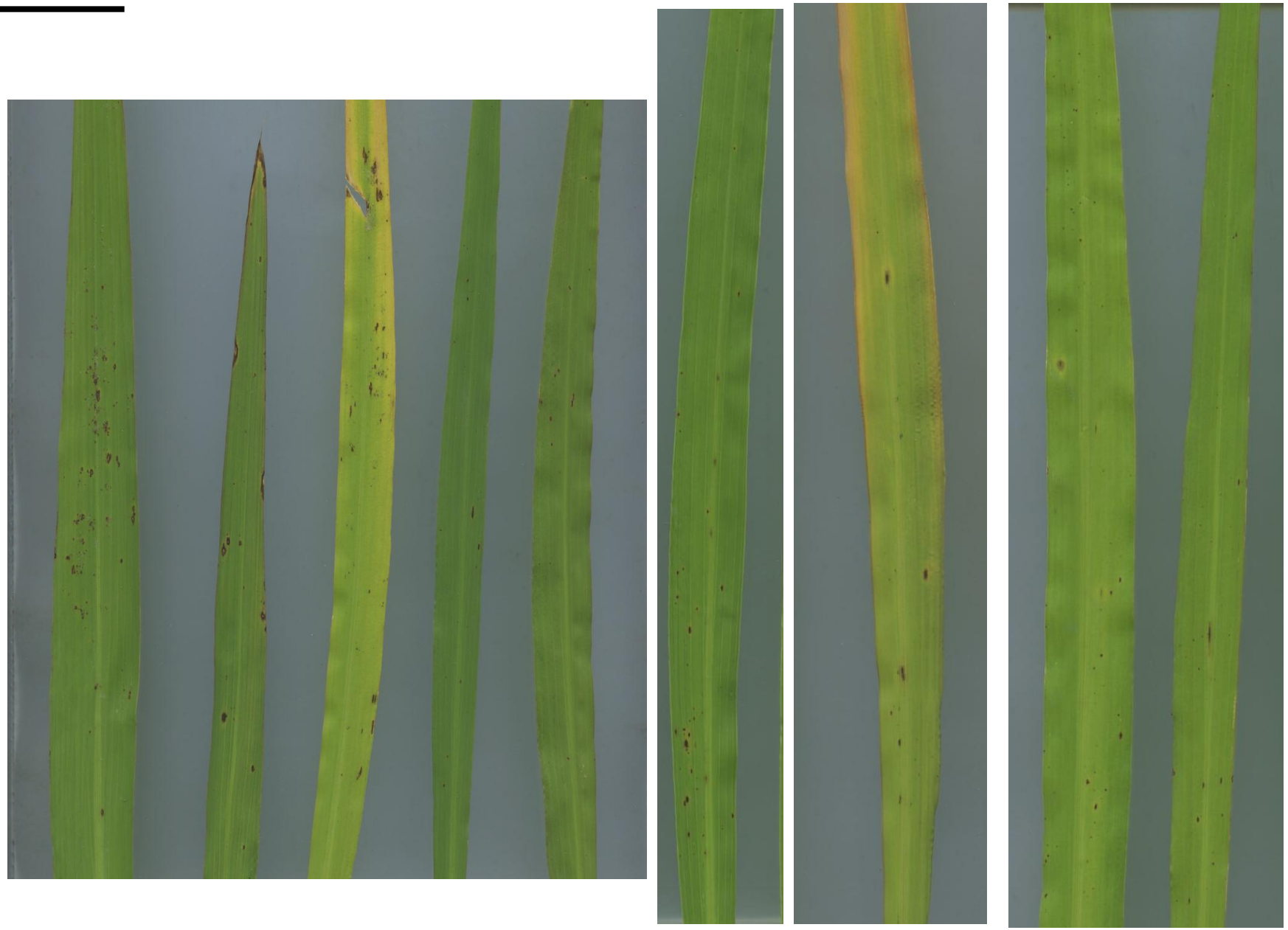
(2)



Author images: Claudia Castell-Miller

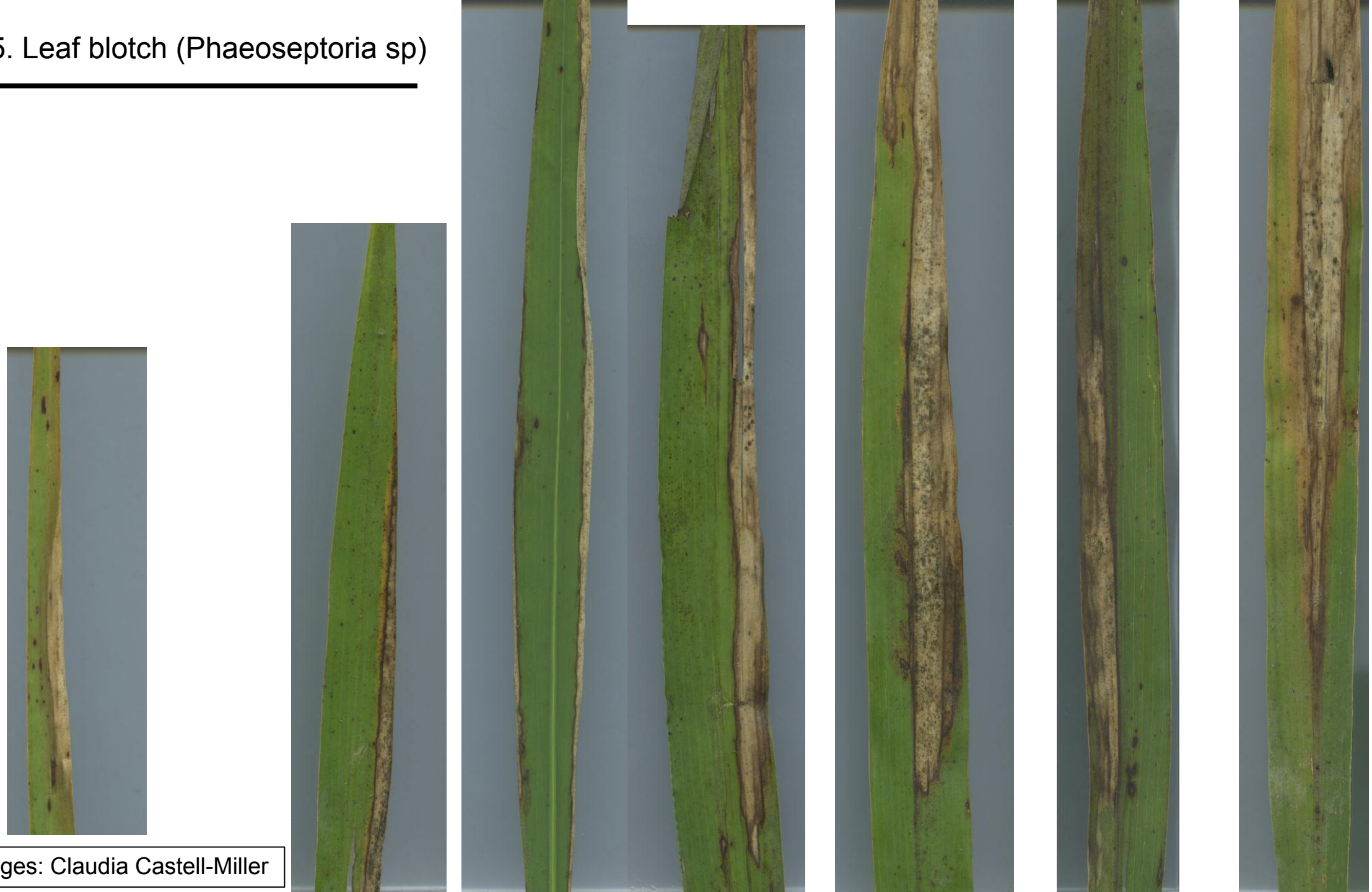
Figure 4. Leaf spot (*Bipolaris zeicola*)

Symptoms 2 weeks after the artificial inoculation in greenhouse



Author images: Claudia Castell-Miller

Figure 5. Leaf blotch (Phaeoseptoria sp)



Author images: Claudia Castell-Miller

Figure 6. Smut (*Eballistra lineata*)



Author images: Claudia Castell-Miller

Figure 7. Johnson lesions (physiological disorder that could be induced by microorganisms)







Author images: Claudia Castell-Miller

Figure 8. Bacteria: *Pseudomonas* spp. (Pictorial Guide)



Authors image: James Percich and Robert Nyvall

Figure 9. Bacteria: *Xanthomonas* sp.



(Pictorial guide)

Authors image: James Percich and Robert Nyvall

Figure 10. Fusarium Head Blight (Scab; *Fusarium* spp.)



(Pictorial guide)

Authors image: James Percich and Robert Nyvall

Figure 11. Ergot (Claviceps *zizaniae)*







Author images: Claudia Castell-Miller

Figure 12. Leaf Sheath and Stem Rot



(Pictorial guide)

Authors image: James Percich and Robert Nyvall